Stefan Girstmair

Curriculum Vitae

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stefangirstmair.github.io

Research Interests

Macroeconomics

- Monetary Economics
- Forecasting
- Bayesian Econometrics

Professional Experience

September 2024–	**Senior Economist**, Bank of Lithuania - Applied Macroeconomic Research Division, Vilnius
January 2024–May 2024	**Consultant**, ECB - European Central Bank - Forecasting and Policy Modelling Division (DGE), Frankfurt am Main
May 2023–September 2023	**PhD-Trainee**, ECB - European Central Bank - Forecasting and Policy Modelling Division (DGE), Frankfurt am Main
October 2018–September 2024	**Graduate Research and Teaching Assistant**, *Chair of International Macroeconomics and Macroeconometrics*, Goethe University, Frankfurt am Main
July 2015	**Intern**, OeNB - Austrian National Bank - Economic Analysis Department, Forecasting, Vienna
July 2014	**Intern**, OeNB - Austrian National Bank - Economic Analysis Department, Forecasting, Vienna

Education

2017–2024	**Ph.D. in Economics**, *Graduate School of Economics, Finance and Management*, Goethe University Thesis title: *"Essays in Macroeconomics"*, Advisors: Prof. Michael Binder and Prof. Volker Wieland
2015–2017	**M.Sc. Economics**, *Institute for Advanced Studies (IHS)* Thesis: *"Unemployment in Austria: Job Losing and Job Finding"*
2012–2015	**B.Sc. Economics**, *University of Vienna* Thesis: *"The Curse of Natural Resources: A Survey and Two Cases"*

Research

LITMAS: The Lithuanian Model for Analysis of Scenarios, *Bank of Lithuania Working Paper, forthcoming*,

Abstract: We develop a large scale open economy DSGE model for policy analysis in Lithuania within the Euro Area (EA) monetary union. The model features two fully specified regions (Lithuania and the rest of the EA) linked through trade flows, a common monetary policy, and a fixed nominal exchange rate. Each region comprises a goods production sector, a construction sector, households, and a fiscal authority. The model incorporates international trade in both consumer and investment goods, with separate pricing mechanisms for each category and region. An energy good imported from outside the euro area serves as an input in goods production, construction, and household consumption. We estimate the model using Lithuanian and EA data, providing an empirically grounded framework for scenario analysis and evaluation of fiscal policy measures in a general equilibrium setting. The detailed fiscal structure and an energy sector integration make the model particularly suited for analysing policy transmission mechanisms through scenario analyses and simulations.

The Effect of New Housing Supply in Structural Models: A Forecasting Performance Evaluation, *ECB Working Paper Series No. 2895*,

Abstract: This paper investigates the importance of including data on new housing supply in Dynamic Stochastic General Equilibrium (DSGE) models in forecasting the Great Financial Crisis (GFC), focusing on the U.S. While existing models have added a financial sector and real estate sector, they have largely overlooked housing supply. I develop an extended DSGE model that includes both the financial sector and endogenous housing supply and show that forecasting accuracy significantly improves when data on new houses is included. Robustness checks confirm the importance of these additions to the model. The findings highlight the necessity of combining model extension and housing supply data for accurate forecasting during economic crises. I identify negative housing demand shocks and escalating adjustment costs as primary drivers of the GFC, propagating into the real economy and accelerating through the financial sector. Additionally, this paper addresses the zero lower bound challenge in modeling forward guidance using a regime change approach.

Determinacy in Multi-Country DSGE Models: The Role of Pricing Paradigms and Economic Openness, *Dynare Working Paper Series No. 82 & Bank of Lithuania Working Paper No. 137/2025*,

Abstract: This paper examines determinacy properties in a multi-country open economy framework, focusing on the impacts of dominant currency pricing (DCP), producer currency pricing (PCP), and local currency pricing (LCP) on monetary policy effectiveness. Utilizing a New Keynesian model with three symmetric economies, each guided by Taylor rules, the study extends the framework of Gopinath et al. (2020) to analyze how these pricing paradigms interact with central bank policies to achieve economic stability. The investigation highlights that higher economic openness amplifies interactions among central banks' policies, complicating the attainment of determinacy. DCP significantly constrains policy parameters ensuring determinacy, particularly in open economies. Conversely, PCP and LCP offer relatively larger determinacy regions, allowing for greater domestic policy control. The findings emphasize the critical role of pricing paradigms and economic openness in formulating effective monetary policies. This study provides essential insights for central banks and policymakers in enhancing global economic stability through tailored policy recommendations based on the chosen pricing paradigm.

A Policy-Relevant Structural Macroeconomic Model for Emerging Market Economies with Illustration for Vietnam, *Together with Michael Binder, Le Van Ha and Anh H. Le*,

Abstract: To conduct policy analysis, for advanced economies usually numerous micro-founded, institutionally relatively detailed, structural macroeconomic models are available. For emerging market economies, in contrast, there tends to be a lack of models of this type that contain the requisite institutional detail to forecast on par with reduced-form time-series models and carry out credible policy analysis. This paper considers as an example of an emerging market economy Vietnam, and introduces a New Keynesian-Dynamic Stochastic General Equilibrium (NK-DSGE) model that depicts Vietnam as an open economy that interacts with the U.S. and the rest of the world. We incorporate various institutional characteristics that we argue are representative for emerging market economies at least in Asia. These include: (i) international trade occurs within the global value chain and under the dominant currency pricing paradigm; (ii) there is a diverse production sector in the domestic economy that comprises privately-owned, state-owned, and foreign direct investment firms facing asymmetric financial frictions; and (iii) monetary policy in part pursues an exchange-rate target. We document that the estimated model can forecast core macroeconomic variables on par (and, at least partially, better) than a state-of-the-art Bayesian Vector Autoregressive model, and thus argue that our model can be fruitfully used for policy analysis. Among the main implications of our model are that while Vietnam's output losses stemming from contractionary domestic monetary policy shocks are larger than prior structural models would suggest, the transmission of foreign shocks to the Vietnamese economy is weaker.

Policy

Indexation, Reserves, and Crisis Resilience: An Assessment of Lithuania's Social Insurance Fund, *Occasional Paper Series, No. 60/2026, Bank of Lithuania*,

Abstract: This paper assesses reserves of Lithuania's social insurance fund (Sodra) under demographic ageing and macroeconomic stress, projecting the budget through 2050 under two pension indexation rules: the current law and a seven-year average rule that allows for reserves to be used for pension expenses. The choice between the rules produces a sharp trade-off between reserve accumulation and pension adequacy. Within the seven-year average baseline, investment and fiscal instruments can ease the demographic pressure but cannot resolve it. Under a Global Financial Crisis-style shock, the indexation rule determines the distribution of crisis costs: reserves absorb the shock under one rule, pensioners under the other.

An Ex-Ante Assessment of the Proposal to Reform the Second Pillar of the Lithuanian Pension System, *Occasional Paper Series, No. 57/2025, Bank of Lithuania*,

Abstract: This document offers an ex-ante assessment of the proposal to allow the withdrawal of funds from the second pillar of the Lithuanian pension system. First, we use a quantitative macroeconomic model to quantify, under alternative scenarios, the potential impact that the withdrawal of Pillar II funds might have on the economy in the medium term. Second, we offer a long-term view of the current pension replacement rates and the consequences that the withdrawal of funds might have for those individuals who decide to opt-out of Pillar II.

Additional Education

June 2024 **Heterogeneous-Agent Macroeconomics**, *Goethe Macro Training School*, Adrien Auclert, Matthew Rognlie, and Ludwig Straub

August 2023 **Machine Learning in Macroeconomics**, *ECB*, Jesus Fernandez-Villaverde

August 2019 **Optimal Fiscal and Monetary Policy**, *Study Center Gerzensee*, Mikhail Golosov

Teaching

Solution, Identification, and Estimation of DSGE Models, *PhD-Level, Prof. Michael Binder*, Teaching Assistant
- Winter 2023/24
- Winter 2022/23
- Winter 2021/22
- Winter 2019/20

Macroeconomics 1 (BMAK), *BSc-Level, Prof. Michael Binder*, Teaching Assistant
- Winter 2023/24
- Winter 2022/23
- Winter 2021/22
- Winter 2020/21
- Winter 2018/19

Software and Databases Skills

Advanced	Matlab, Dynare, Office, Git
Proficient	R, Stata, Julia

Languages

German	Native
English	Excellent command
French	Beginner
Korean	Beginner

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